

```
> task_mtcars
<TaskRegr:mtcars> (32 x 11): Motor Trends
* Target: mpg
* Properties: -
* Features (10):
- dbl (10): am, carb, cyl, disp, drat, gear, hp, qsec, vs,
wt
```

The boston_housing data set contains information on various houses in Boston such as the proportion of non-retail business acres per town (indus), the percentage of lower status of the population (lstat), index of accessibility to radial highways (rad), and the average number of rooms per dwelling (rm), etc.

```
> task_boston_housing = tsk("boston_housing")

> task_boston_housing
<TaskRegr:boston_housing> (506 x 19): Boston Housing Prices
* Target: medv
* Properties: -
* Features (18):
- dbl (13): age, b, cmedv, crim, dis, indus, lat, lon,
lstat, nox,
ptratio, rm, zn
- int (3): rad, tax, tract
- fct (2): chas, town
```

The data.table function

The `data.table()` function is implemented in the `data.table` package for storing tabular data. It is an alternative to R's `data.frame`, and provides more functionality and better performance. For example:

```
> dt = as.data.table(task_boston_housing)

> dt

      medv age      b chas cmedv   crim   dis indus   lat
lon lstat
1: 24.0 65.2 396.90    0  24.0 0.00632 4.0900  2.31 42.2550
-70.9550  4.98
2: 21.6 78.9 396.90    0  21.6 0.02731 4.9671  7.07 42.2875
-70.9500  9.14
3: 34.7 61.1 392.83    0  34.7 0.02729 4.9671  7.07 42.2830
-70.9360  4.03
4: 33.4 45.8 394.63    0  33.4 0.03237 6.0622  2.18 42.2930
-70.9280  2.94
5: 36.2 54.2 396.90    0  36.2 0.06905 6.0622  2.18 42.2980
-70.9220  5.33
...
```

The as_task_regr function

A regression task needs to be created using the `as_task_regr()` function for a given `data.frame` type object. The `subset()` function is used to retrieve selective data from the data set. In this example, the 'cylinder' and 'gear' columns from the `mtcars` data set are selected, and the gear column is chosen as the prediction field:

```
> mtcars_set = subset(mtcars, select = c("cyl", "gear"))

> str(mtcars_set)
'data.frame':  32 obs. of  2 variables:
 $ cyl : num  6 6 4 6 8 6 8 4 4 6 ...
 $ gear: num  4 4 4 3 3 3 3 4 4 4 ...

> task_mtcars = as_task_regr(mtcars_set, target = "gear", id
= "car")

> task_mtcars
<TaskRegr:car> (32 x 2)
* Target: gear
* Properties: -
* Features (1):
- dbl (1): cyl
```

The head function

The first few entries of the data set can be viewed using the `head()` function as illustrated:

```
> task_mtcars$head()
      gear cyl
1:      4   6
2:      4   6
3:      4   4
4:      3   6
5:      3   8
6:      3   6
```

The summary function

The `summary()` function can also be used on the data set to display the quartiles, minimum, maximum, median and mean values as illustrated here:

```
> summary(as.data.table(task_mtcars))

      gear      cyl
Min.   :3.000  Min.   :4.000
1st Qu.:3.000  1st Qu.:4.000
Median :4.000  Median :6.000
Mean   :3.688  Mean   :6.188
3rd Qu.:4.000  3rd Qu.:8.000
Max.   :5.000  Max.   :8.000
```